

Calculus and Analytical Geometry

The Limit of a Function

Dr. Aamir Alaud Din

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Objectives

After preparing this topic, you should be able

1. To estimate and interpret the limit of a function numerically and graphically as the independent variable approaches a particular value.
2. To understand left-hand and right-hand limits and use them to determine whether a two-sided limit exists.
3. To identify the different ways in which a limit can fail to exist and recognize why numerical calculations and graphs can sometimes give misleading conclusions.
4. To understand infinite limits and use one-sided behavior to identify vertical asymptotes of functions.

The Why Section

Objective 1: Why Study Limits Numerically and Graphically?

- In engineering, we frequently need to understand what happens to a physical quantity when another quantity gets very close to a particular value.
- For example, suppose the displacement of a vibrating mechanical component is described by a function $s(t)$, and we want to know what happens to its velocity as the time t approaches a particular instant.
- At first, we may not have an algebraic method for calculating the required value directly.
- We can instead investigate values of the function for numbers increasingly close to the point of interest and observe the corresponding behavior.
- Tables provide numerical evidence, while graphs allow us to see the overall behavior visually.
- Therefore, numerical and graphical methods provide an intuitive first approach to the concept of a limit.
- The following type of mechanical-engineering problem can be handled by studying the first objective.
- A sensor records the response of a mechanical system by a function $R(t)$. The system behaves differently at an exact operating time, but we want to determine what value the response approaches as t gets closer and closer to that time.
- We will be able to estimate the required value from a table and graph.

Objective 2: Why Study One-Sided Limits?

- In many physical systems, behavior may be different immediately before and immediately after a particular point.

- For example, a mechanical mechanism may change its operating mode when a control parameter reaches a critical value.
- The response approaching the critical value from below may therefore be different from the response approaching it from above.
- A two-sided limit can exist only when the function approaches the same value from both directions.
- Therefore, left-hand and right-hand limits allow us to examine the behavior on each side separately.
- This idea is especially useful for piecewise-defined functions, switching mechanisms, and engineering models involving sudden changes.
- The following type of problem can be handled by studying one-sided limits.
- A mechanical control system changes its operating rule when the input reaches a critical setting c .
- We want to determine whether the system response approaches the same value as the setting approaches c from below and from above.

Objective 3: Why Study How a Limit Can Fail to Exist?

- It is not always true that a function approaches a single number as x approaches a particular value.
- A limit can fail to exist because the left-hand and right-hand limits are different.
- A limit can also fail to exist because the function oscillates indefinitely or because its values become arbitrarily large.
- Recognizing these possibilities prevents us from making an incorrect conclusion simply by examining a few numerical values.
- This is important in engineering because oscillatory behavior and rapidly increasing responses can indicate important characteristics of a physical system.
- For example, a mathematical model may oscillate increasingly rapidly near a particular operating condition.
- A few calculator values might suggest a particular limit even though the function does not actually approach a single number.
- These situations can be understood by studying the different ways in which a limit can fail to exist.

Objective 4: Why Study Infinite Limits and Vertical Asymptotes?

- In some engineering models, a quantity can become extremely large as an input approaches a particular value.
- For example, a mathematical model of a mechanical response may contain a denominator that becomes very small near a critical parameter value.
- The resulting function may increase without bound as the parameter approaches that value.
- Such behavior is described using an infinite limit.
- A vertical asymptote provides a geometric way to describe this behavior.
- Understanding infinite limits and vertical asymptotes helps us recognize values at which a mathematical model becomes unbounded and therefore requires special interpretation.
- The following type of engineering situation can be analyzed using this idea.

- A simplified mechanical model contains a response function whose denominator approaches zero when an operating parameter reaches a critical value.
- We want to determine what happens to the response near that value and whether a vertical asymptote occurs.

Objective 1: Finding Limits Numerically and Graphically

Investigating a Limit Numerically

- We begin by considering the function

$$f(x) = \frac{x - 1}{x^2 - 1}$$

- We want to investigate the behavior of $f(x)$ for values of x close to 1 but not equal to 1.
- A table of values from both sides of 1 gives

Table 1. Values of x and corresponding $f(x)$ values.

$x < 1$	$f(x)$	$x > 1$	$f(x)$
0.5	0.666667	1.5	0.400000
0.9	0.526316	1.1	0.476190
0.99	0.502513	1.01	0.497512
0.999	0.500250	1.001	0.499750
0.9999	0.500025	1.0001	0.499975

- As x approaches 1 from either side, the values of $f(x)$ approach 0.5.
- Therefore, we estimate that

$$\lim_{x \rightarrow 1} \frac{x - 1}{x^2 - 1} = 0.5$$

- Notice that the question concerns what happens **near** $x = 1$, not necessarily what happens at $x = 1$ itself.

Investigating a Limit Graphically

- The same function can be investigated by examining its graph.

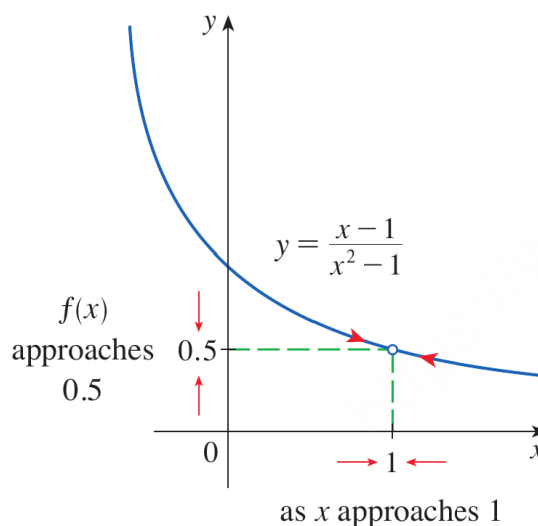


Figure 1. The graph of $f(x) = \frac{x-1}{x^2-1}$ showing $f(x)$ approaching 0.5 as x approaches 1.

- The graph also suggests that the values of $f(x)$ approach 0.5 as x approaches 1.
- Thus, numerical tables and graphs give two complementary ways of investigating a limit.

Definition: Intuitive Definition of a Limit

Suppose $f(x)$ is defined when x is near a , possibly except at a itself.

We write the following limit and say that **the limit of $f(x)$ as x approaches a is L** if the values of $f(x)$ can be made arbitrarily close to L by taking x sufficiently close to a on either side of a , but not equal to a .

$$\lim_{x \rightarrow a} f(x) = L$$

Roughly speaking, the values of $f(x)$ approach L as x approaches a .

The function does not even have to be defined at $x = a$ for the limit to exist.

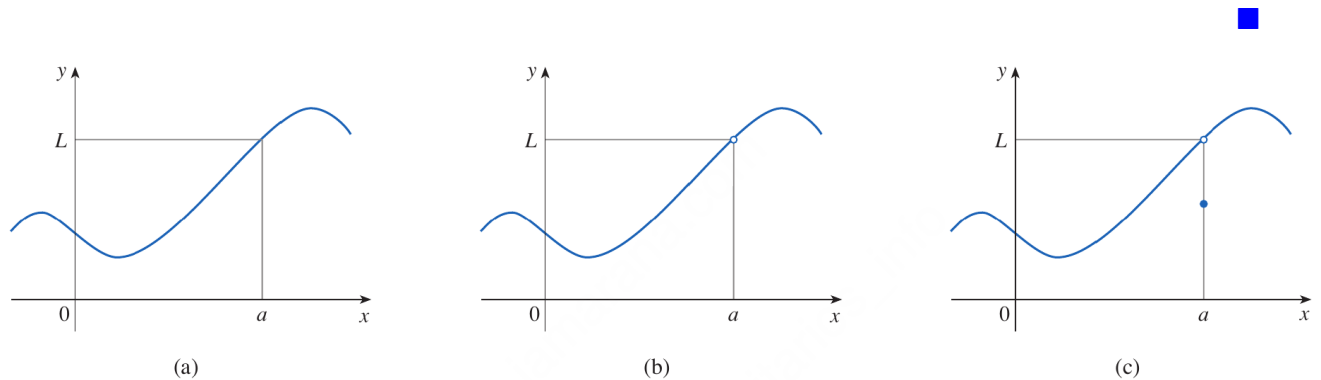


Figure 2. Graphs illustrating that $\lim_{x \rightarrow a} f(x) = L$ can exist even when the value of $f(a)$ is different or undefined.

- The three graphs illustrate an important point: the behavior of $f(x)$ near a determines the limit, not necessarily the value of $f(a)$.
- An alternative notation is

$$f(x) \rightarrow L \quad \text{as} \quad x \rightarrow a$$

- This is usually read as " $f(x)$ approaches L as x approaches a ."

Example 1

Estimate the value of

$$\lim_{t \rightarrow 0} \frac{\sqrt{t^2 + 9} - 3}{t^2}$$

Solution

We construct a table of values for t near 0.

Table 2. The values of t and corresponding function.

t	$\frac{\sqrt{t^2 + 9} - 3}{t^2}$
± 1.0	0.162277...
± 0.5	0.165525...
± 0.1	0.166620...
± 0.05	0.166655...
± 0.01	0.166666...

The values appear to approach 0.166666...

Therefore, our initial estimate is

$$\lim_{t \rightarrow 0} \frac{\sqrt{t^2 + 9} - 3}{t^2} = \frac{1}{6}$$

If we use extremely small values of t , however, a calculator may give misleading results.

For example, direct calculation can eventually produce values such as 0.000000 because of rounding errors.

This does not mean that the limit is 0.

The actual limit is

$$\frac{1}{6}$$

This example shows that numerical calculations are useful for estimating limits, but they must be used carefully.

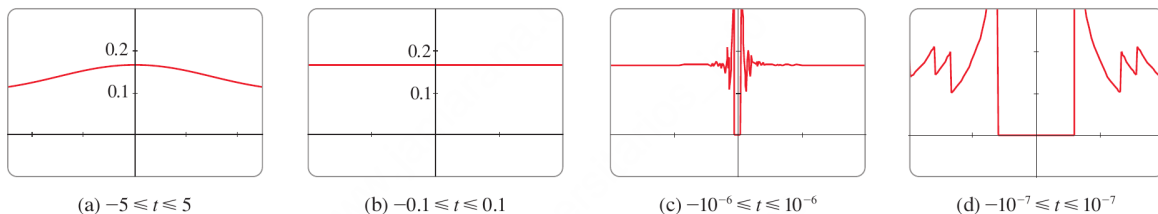


Figure 3. The graph of $y = \frac{\sin x}{x}$ near $x = 0$.

The following Python programs shows that the behaviour of the function is captured with small step increments (in the order of 1×10^{-6}) in the independent variable near the limiting points.

```
import numpy as np
import matplotlib.pyplot as plt

def f(t):
    return (np.sqrt(t**2 + 9) - 3) / t**2

t_values = np.linspace(-5, 5, 1000)
plt.plot(t_values, f(t_values), linewidth=2, color='k')
plt.xticks(fontsize=12, fontweight='bold')
plt.yticks(fontsize=12, fontweight='bold')
plt.xlabel('t', fontsize=14, fontweight='bold')
plt.ylabel('f(t)', fontsize=14, fontweight='bold')
plt.title("Big Jumps", fontsize=16, fontweight='bold')
plt.grid(True)
```

```

plt.show()

t_values = np.linspace(-0.01, 0.01, 1000)
plt.plot(t_values, f(t_values), linewidth=2, color='k')
plt.xticks(fontsize=12, fontweight='bold')
plt.yticks(fontsize=12, fontweight='bold')
plt.xlabel('t', fontsize=14, fontweight='bold')
plt.ylabel('f(t)', fontsize=14, fontweight='bold')
plt.title("Very Small Jumps", fontsize=16, fontweight='bold')
plt.grid(True)
plt.show()

t_values = np.linspace(-0.000001, 0.000001, 1000)
plt.plot(t_values, f(t_values), linewidth=2, color='k')
plt.xticks(fontsize=12, fontweight='bold')
plt.yticks(fontsize=12, fontweight='bold')
plt.xlabel('t', fontsize=14, fontweight='bold')
plt.ylabel('f(t)', fontsize=14, fontweight='bold')
plt.title("Microscopic Jumps", fontsize=16, fontweight='bold')
plt.grid(True)
plt.show()

t_values = np.linspace(-0.0000001, 0.0000001, 1000)
plt.plot(t_values, f(t_values), linewidth=2, color='k')
plt.xticks(fontsize=12, fontweight='bold')
plt.yticks(fontsize=12, fontweight='bold')
plt.xlabel('t', fontsize=14, fontweight='bold')
plt.ylabel('f(t)', fontsize=14, fontweight='bold')
plt.title("Ten Times Smaller Microscopic Jumps", fontsize=16, fontweight='bold')
plt.grid(True)
plt.show()

t_values = np.linspace(-0.00000001, 0.00000001, 1000)
plt.plot(t_values, f(t_values), linewidth=2, color='k')
plt.xticks(fontsize=12, fontweight='bold')
plt.yticks(fontsize=12, fontweight='bold')
plt.xlabel('t', fontsize=14, fontweight='bold')
plt.ylabel('f(t)', fontsize=14, fontweight='bold')
plt.title("100 Times Stronger Microscopic Jumps", fontsize=16, fontweight='bold')
plt.grid(True)
plt.show()

```

Example 2

Guess the value of

$$\lim_{x \rightarrow 0} \frac{\sin x}{x}$$

Solution

The function $f(x) = \frac{\sin x}{x}$ is not defined at $x = 0$.

We calculate values for x close to 0.

Table 3. Values of x and corresponding function values.

x	$\frac{\sin x}{x}$
± 1.0	0.84147098
± 0.5	0.95885108
± 0.4	0.97354586
± 0.3	0.98506736
± 0.2	0.99334665
± 0.1	0.99833417
± 0.05	0.99958339
± 0.01	0.99998333
± 0.005	0.99999583
± 0.001	0.99999983

The values approach 1 as x approaches 0.

Therefore, we guess that

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

This result is correct and will later be established using a geometric argument.

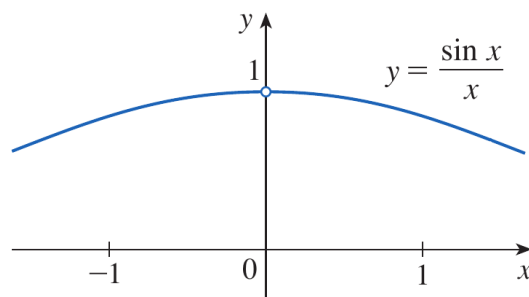


Figure 4. The graph of $\frac{\sin x}{x}$ function.



Example 3

Find

$$\lim_{x \rightarrow 0} \left(x^3 + \frac{\cos 5x}{10,000} \right)$$

Solution

We first calculate values of the function for several values of x near 0.

Table 4. Values of x with bigger increments and corresponding function values.

x	$x^3 + \frac{\cos 5x}{10,000}$
1	1.000028
0.5	0.124920
0.1	0.001088
0.05	0.000222
0.01	0.000101

For values even closer to zero we obtain

Table 5. Values of x with shorter increments and corresponding function values.

x	$x^3 + \frac{\cos 5x}{10,000}$
0.005	0.00010009
0.001	0.00010000

The values approach 0.0001.

Therefore,

$$\lim_{x \rightarrow 0} \left(x^3 + \frac{\cos 5x}{10,000} \right) = 0.0001$$



Limits and Technology

- Calculators and computer algebra systems can be useful when investigating limits numerically.
- However, numerical calculations can sometimes produce false impressions because of rounding and finite numerical precision.
- Therefore, tables and graphs should be viewed as tools for **estimating and understanding** a limit rather than as proof that a limit has a particular value.

Quick Check

Question 1

Use the following information to estimate the limit.

Table 6. Value of x and corresponding values of $f(x)$.

x	$f(x)$
0.9	2.1
0.99	2.01
0.999	2.001
1.001	1.999
1.01	1.99

What value does $f(x)$ appear to approach as $x \rightarrow 1$?

Question 2

If $f(x)$ approaches 5 as x approaches 3, but $f(3) = 10$, what is $\lim_{x \rightarrow 3} f(x)$?



Objective 2: One-Sided Limits

Approaching from One Side

- Sometimes we are interested in what happens as x approaches a from only one direction.
- The following notation means that we consider values of x less than a and approach a from the left.

$$x \rightarrow a^-$$

- Similarly, the following notation means means that we consider values of x greater than a and approach a from the right.

$$x \rightarrow a^+$$

The Heaviside Function

- Consider the Heaviside function H defined by

$$H(t) = \begin{cases} 0 & \text{if } t < 0 \\ 1 & \text{if } t \geq 0 \end{cases}$$

- This function can describe an electric current that is switched on at time $t = 0$.
- As t approaches 0 from the left, $H(t)$ approaches 0.
- As t approaches 0 from the right, $H(t)$ approaches 1.
- Since the two values are different, there is no single two-sided limit at $t = 0$.
- We write

$$\lim_{t \rightarrow 0^-} H(t) = 0 \quad \text{and} \quad \lim_{t \rightarrow 0^+} H(t) = 1$$

- These are called the **left-hand limit** and **right-hand limit**, respectively.

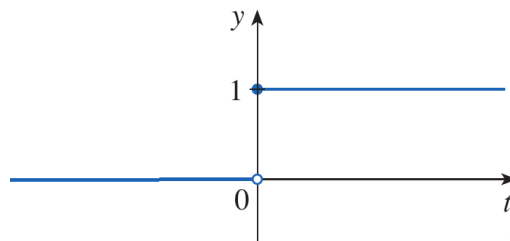


Figure 5. One-sided approaches to a from the left and from the right.

Definition: Intuitive Definitions of One-Sided Limits

We write the following limit when the values of $f(x)$ can be made arbitrarily close to L by restricting x to be sufficiently close to a with $x < a$.

$$\lim_{x \rightarrow a^-} f(x) = L$$

Similarly, the following limit means means that the values of $f(x)$ can be made arbitrarily close to L by restricting x to be sufficiently close to a with $x > a$.

$$\lim_{x \rightarrow a^+} f(x) = L$$

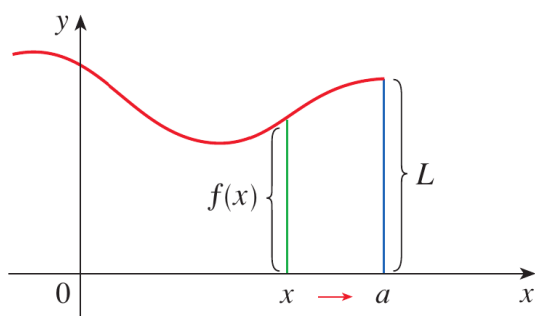


- Thus, the only difference between one-sided and ordinary limits is the direction from which x approaches a .
- For example, $x \rightarrow 5^-$ means that we consider only values $x < 5$, while $x \rightarrow 5^+$ means that we consider only values $x > 5$.

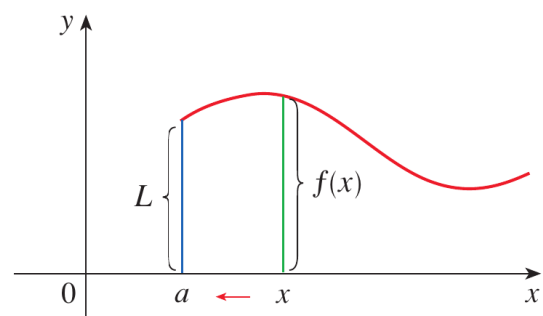
Relationship Between One-Sided and Two-Sided Limits

- A two-sided limit exists only if both one-sided limits exist and are equal.
- Therefore,

$$\lim_{x \rightarrow a} f(x) = L \iff \lim_{x \rightarrow a^-} f(x) = L \text{ and } \lim_{x \rightarrow a^+} f(x) = L$$



(a) $\lim_{x \rightarrow a^-} f(x) = L$



(b) $\lim_{x \rightarrow a^+} f(x) = L$

Figure 6. Graph of the function g used to determine its one-sided and two-sided limits.

Example 4

The graph of a function g is shown in Figure 7. Use the graph to state the values, if they exist, of the following

(a) $\lim_{x \rightarrow 2^-} g(x)$

(b) $\lim_{x \rightarrow 2^+} g(x)$

(c) $\lim_{x \rightarrow 2} g(x)$

(d) $\lim_{x \rightarrow 5^-} g(x)$

(e) $\lim_{x \rightarrow 5^+} g(x)$

(f) $\lim_{x \rightarrow 5} g(x)$



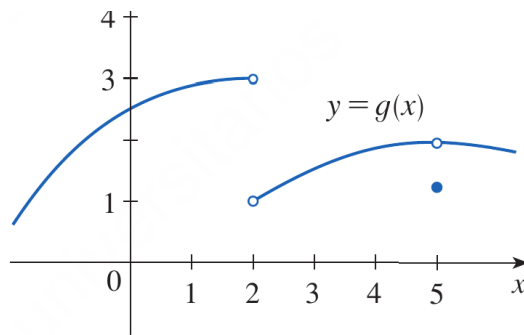


Figure 7. The graph of a function g .

Solution

From the graph, the values of $g(x)$ approach 3 as x approaches 2 from the left.

$$\lim_{x \rightarrow 2^-} g(x) = 3$$

As x approaches 2 from the right, the values approach 1.

$$\lim_{x \rightarrow 2^+} g(x) = 1$$

Because the left-hand and right-hand limits are different, the two-sided limit does not exist.

$$\lim_{x \rightarrow 2} g(x) \text{ does not exist}$$

At $x = 5$, both one-sided limits approach 2.

$$\lim_{x \rightarrow 5^-} g(x) = 2$$

$$\lim_{x \rightarrow 5^+} g(x) = 2$$

Since these one-sided limits are equal,

$$\lim_{x \rightarrow 5} g(x) = 2$$

Notice that the graph has $g(5) \neq 2$.

Again, this demonstrates that the value of the function at $x = a$ does not by itself determine the limit.



Quick Check

Question 1

Suppose

$$\lim_{x \rightarrow 3^-} f(x) = 4 \quad \text{and} \quad \lim_{x \rightarrow 3^+} f(x) = 4$$

What is the two-sided limit?

Question 2

Suppose

$$\lim_{x \rightarrow 2^-} f(x) = 5 \quad \text{and} \quad \lim_{x \rightarrow 2^+} f(x) = 7$$

Does $\lim_{x \rightarrow 2} f(x)$ exist?



Objective 3: How Can a Limit Fail to Exist?

Different One-Sided Limits

- The first way a limit can fail to exist is when the left-hand and right-hand limits are different.
- Example 4 demonstrated this situation at $x = 2$.
- Since

$$\lim_{x \rightarrow 2^-} g(x) = 3$$

- While

$$\lim_{x \rightarrow 2^+} g(x) = 1$$

- There is no single number approached by $g(x)$ from both sides.
- Therefore,

$\lim_{x \rightarrow 2} g(x) \text{ does not exist}$
--

Oscillation

- A second way a limit can fail to exist is when the function oscillates between different values and does not approach a single number.
- The similar behaviour was observed in Example 1.

Example 5

Investigate

$$\lim_{x \rightarrow 0} \sin \frac{\pi}{x}$$

Solution

The function $f(x) = \sin(\pi/x)$ is undefined at $x = 0$.

If we choose values such as

$$x = 1, \quad \frac{1}{2}, \quad \frac{1}{3}, \quad \frac{1}{10}, \quad \frac{1}{100}$$

then

$$\sin \frac{\pi}{x}$$

gives values such as 0.

However, this information is misleading.

For every positive integer n ,

$$f\left(\frac{1}{n}\right) = \sin(n\pi) = 0$$

But there are also sequences of values approaching zero for which the function equals 1 or -1 .

For example, if

$$x = \frac{2}{4n+1}$$

then

$$\frac{\pi}{x} = \frac{(4n+1)\pi}{2}$$

and the sine is 1.

Similarly, other values approaching zero produce -1 .

Thus, as x approaches 0, the function continues to oscillate between -1 and 1.

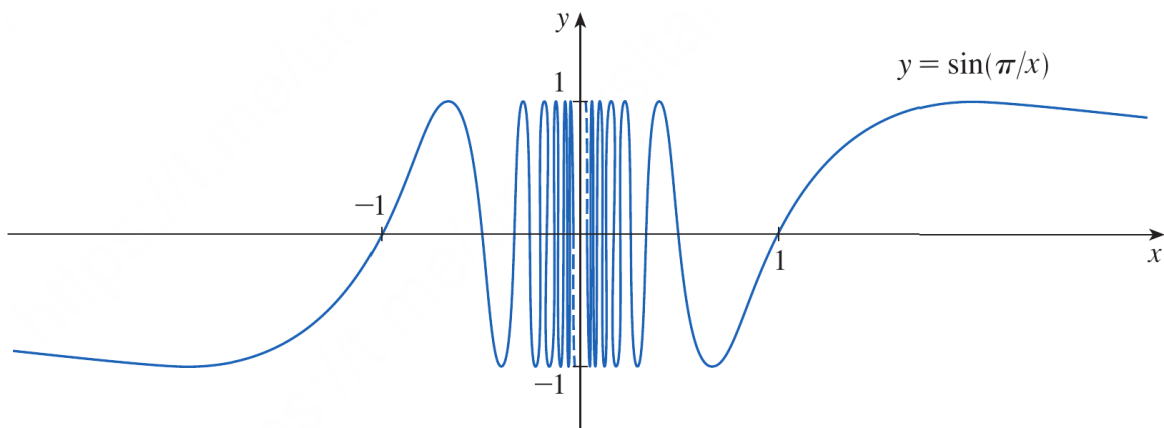


Figure 8.1. Graph of $y = \sin(\pi/x)$ showing increasingly rapid oscillation near $x = 0$.

Because the function does not approach a single fixed number,

$$\lim_{x \rightarrow 0} \sin \frac{\pi}{x} \text{ does not exist}$$

This example shows why checking only a few calculator values can lead to a wrong conclusion.

The following Python program shows this behaviour in depth.

```
import numpy as np
import matplotlib.pyplot as plt

def f(x):
    return np.sin(np.pi / x)

x_left = np.linspace(-1, -0.0003, 10000)
x_right = np.linspace(0.0003, 1, 10000)
```

```

y_left = f(x_left)
y_right = f(x_right)

plt.plot(x_left, y_left, linewidth=2)
plt.plot(x_right, y_right, linewidth=2)
plt.xticks(fontsize=12, fontweight='bold')
plt.yticks(fontsize=12, fontweight='bold')
plt.xlabel('x', fontsize=14, fontweight='bold')
plt.ylabel('f(x)', fontsize=14, fontweight='bold')
plt.grid(True)
plt.show()

```

The output of this program is shown below.

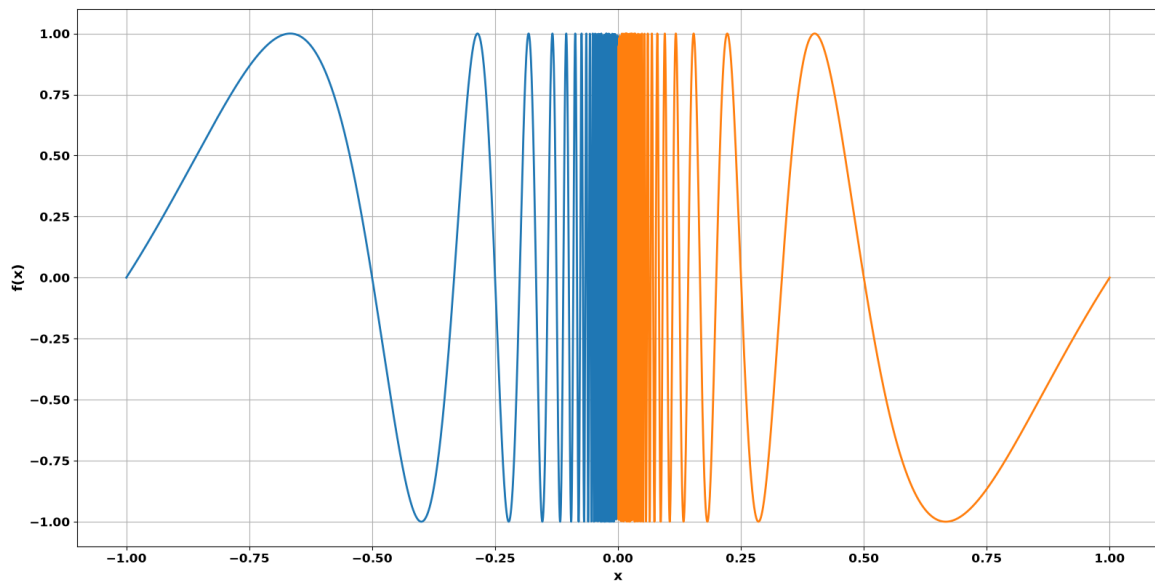


Figure 8.2. The graph of $\sin \frac{\pi}{x}$ for $-1 \leq x \leq 1$.

Unbounded Behavior

- A third situation occurs when the function values become arbitrarily large in magnitude as x approaches a particular number.

- Consider

$$f(x) = \frac{1}{x^2}$$

- As x approaches 0, the denominator becomes very small and positive.
- Consequently, $1/x^2$ becomes larger and larger.

Example 6

Find

$$\lim_{x \rightarrow 0} \frac{1}{x^2}$$

if it exists

Solution

Some values of the function are

Table 7. Values of x and corresponding function values.

x	$\frac{1}{x^2}$
± 1	1
± 0.5	4
± 0.2	25
± 0.1	100
± 0.05	400
± 0.01	10,000
± 0.001	1,000,000

The values become larger and larger as x approaches 0.

Therefore, the function does not approach a finite number.

We express this behavior by writing

$$\lim_{x \rightarrow 0} \frac{1}{x^2} = \infty$$

The symbol ∞ does **not** represent a number.

It indicates that the function values become arbitrarily large.

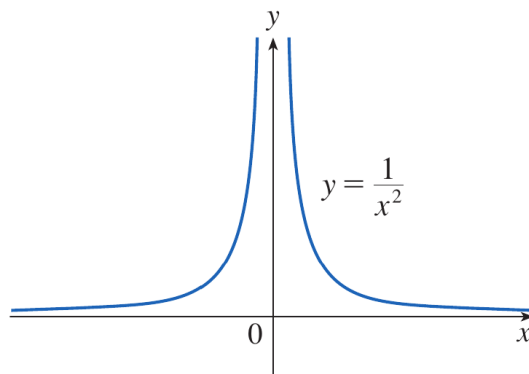


Figure 9. Graph of $y = \frac{1}{x^2}$ showing unbounded behavior near $x = 0$.

Infinite Limits

- In general, the following limit means that $f(x)$ can be made arbitrarily large by taking x sufficiently close to a , but not equal to a .

$$\lim_{x \rightarrow a} f(x) = \infty$$

- Similarly, below limit means that $f(x)$ becomes arbitrarily large in the negative direction as x approaches a .

$$\lim_{x \rightarrow a} f(x) = -\infty$$

- These expressions describe behavior; they do not mean that the limit exists as an ordinary real number.

Another Form of Unbounded Behavior

- Consider

$$f(x) = -\frac{1}{x^2}$$

- As x approaches 0, the function values become arbitrarily large in the negative direction.
- Therefore,

$$\lim_{x \rightarrow 0} \left(-\frac{1}{x^2} \right) = -\infty$$

Quick Check

Question 1

Does the following limit exist?

$$\lim_{x \rightarrow 0} \sin \frac{1}{x}$$

Question 2

What type of behavior is represented by

$$\lim_{x \rightarrow 2} f(x) = \infty$$



Objective 4: Infinite Limits; Vertical Asymptotes

Vertical Asymptotes

- The unbounded behavior of a function is closely related to the idea of a vertical asymptote.
- A vertical line $x = a$ is called a **vertical asymptote** of the curve $y = f(x)$ if at least one of the following one-sided or two-sided limits is infinite:

$$\lim_{x \rightarrow a^-} f(x) = \infty$$

$$\lim_{x \rightarrow a^+} f(x) = \infty$$

$$\lim_{x \rightarrow a} f(x) = \infty$$

$$\lim_{x \rightarrow a^-} f(x) = -\infty$$

$$\lim_{x \rightarrow a^+} f(x) = -\infty$$

$$\lim_{x \rightarrow a} f(x) = -\infty$$

- Thus, a vertical asymptote describes a value of x near which the function becomes unbounded.

Determining Vertical Asymptotes Using One-Sided Limits

Example 7

Does the curve

$$y = \frac{2x}{x - 3}$$

have a vertical asymptote?

Solution

A potential vertical asymptote occurs where the denominator is zero.

We solve

$$x - 3 = 0$$

giving

$$x = 3$$

We therefore investigate the behavior from both sides.

If x is close to 3 but greater than 3, then $x - 3$ is a small positive number, while $2x$ is close to 6.

Therefore, the quotient becomes very large and positive.

$$\lim_{x \rightarrow 3^+} \frac{2x}{x - 3} = \infty$$

If x is close to 3 but less than 3, then $x - 3$ is a small negative number, while $2x$ is still positive.

Therefore, the quotient becomes very large and negative.

$$\lim_{x \rightarrow 3^-} \frac{2x}{x - 3} = -\infty$$

Hence, the line

$$x = 3$$

is a vertical asymptote.

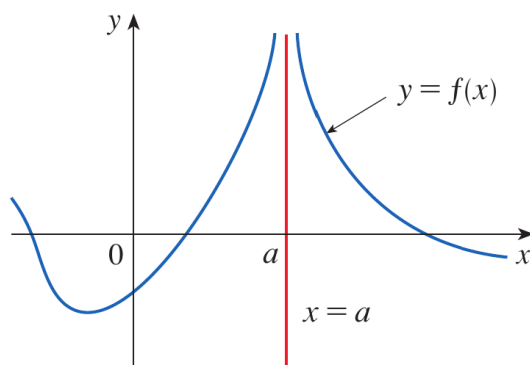


Figure 10. Graph of $y = \frac{2x}{x - 3}$ showing the vertical asymptote $x = 3$.

Important Note

Neither of the two-sided infinite limits is an ordinary finite limit.

In this example, because the function approaches $+\infty$ from one side and $-\infty$ from the other, it is more appropriate to describe the behavior using the two one-sided limits.



Vertical Asymptotes of the Tangent Function

Example 8

Find the vertical asymptotes of $f(x) = \tan x$

Solution

Since

$$\tan x = \frac{\sin x}{\cos x}$$

potential vertical asymptotes occur where

$$\cos x = 0$$

The solutions are

$$x = \frac{\pi}{2} + n\pi$$

where n is any integer.

For example, near $x = \pi/2$,

$$\lim_{x \rightarrow (\pi/2)^-} \tan x = \infty$$

and

$$\lim_{x \rightarrow (\pi/2)^+} \tan x = -\infty$$

Therefore,

$$x = \frac{\pi}{2} + n\pi, \quad n \in \mathbb{Z}$$

are the vertical asymptotes of $y = \tan x$.

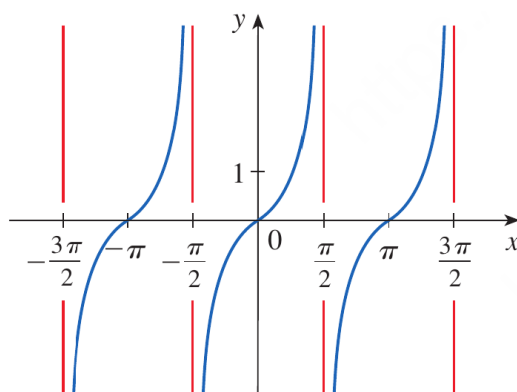


Figure 11. Graph of $y = \tan x$ showing its vertical asymptotes.

Vertical Asymptote of the Natural Logarithm

- Another example of a function with a vertical asymptote is the natural logarithm

$$y = \ln x$$

- As x approaches 0 from the right, the logarithm decreases without bound.

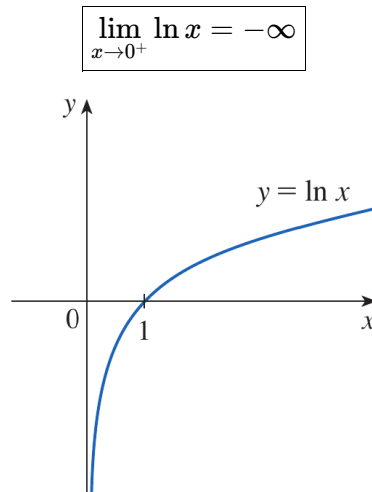


Figure 12. Graph of $y = \ln x$ showing the vertical asymptote $x = 0$.

- Therefore, the line $x = 0$ is a vertical asymptote of $y = \ln x$.
- The same idea applies to $y = \log_b x$ when $b > 1$.

Quick Check

Question 1

Determine the vertical asymptote of

$$f(x) = \frac{1}{x - 4}$$

Question 2

Determine the vertical asymptotes of

$$f(x) = \tan x$$



Scenario Problem

Problem

A simplified mechanical model describes the response of a component near a critical operating parameter $x = 3$ by

$$R(x) = \frac{2x}{x - 3}$$

The parameter x represents a normalized operating condition, and $R(x)$ represents the corresponding response of the component.

Determine what happens to the response as the operating condition approaches the critical value $x = 3$ from the left and from the right, and determine whether $x = 3$ is a vertical asymptote.

Solution

We first consider values of x slightly less than 3.

For example, if $x = 2.9$, then

$$R(2.9) = \frac{2(2.9)}{2.9 - 3}$$

$$R(2.9) = \frac{5.8}{-0.1} = -58$$

If $x = 2.99$, then

$$R(2.99) = \frac{5.98}{-0.01} = -598$$

The response becomes increasingly negative as x approaches 3 from the left.

Therefore,

$$\lim_{x \rightarrow 3^-} R(x) = -\infty$$

Now consider values slightly greater than 3.

If $x = 3.1$, then

$$R(3.1) = \frac{6.2}{0.1} = 62$$

If $x = 3.01$, then

$$R(3.01) = \frac{6.02}{0.01} = 602$$

The response becomes increasingly positive as x approaches 3 from the right.

Therefore,

$$\lim_{x \rightarrow 3^+} R(x) = \infty$$

Since the response becomes unbounded near $x = 3$, the line

$$x = 3$$

is a vertical asymptote.

Thus, the mathematical model predicts extremely large positive or negative responses as the operating condition approaches the critical value from the corresponding side.



Summary

- A **limit** describes the value that $f(x)$ approaches as x gets closer and closer to a particular number.
- Numerical tables can be used to estimate a limit by evaluating the function at values increasingly close to the point of interest.
- Graphs provide a visual way to observe the behavior of a function near the point being approached.
- The intuitive meaning of a limit is expressed by

$$\lim_{x \rightarrow a} f(x) = L$$

- The value of $f(a)$ does not necessarily determine the value of the limit, and the function does not even have to be defined at a .
- A **left-hand limit** considers values of x approaching a from values less than a .

$$\lim_{x \rightarrow a^-} f(x)$$

- A **right-hand limit** considers values of x approaching a from values greater than a .

$$\lim_{x \rightarrow a^+} f(x)$$

- A two-sided limit exists only when the two one-sided limits exist and are equal.

$$\lim_{x \rightarrow a} f(x) = L \iff \lim_{x \rightarrow a^-} f(x) = L \quad \text{and} \quad \lim_{x \rightarrow a^+} f(x) = L$$

- A limit can fail to exist when the left-hand and right-hand limits are different.
- A limit can also fail to exist when the function oscillates indefinitely, as in

$$\lim_{x \rightarrow 0} \sin \frac{\pi}{x} \text{ does not exist}$$

- A function can also become arbitrarily large or negative as x approaches a point.
- The following notation describes values of $f(x)$ that increase without bound and does not mean that ∞ is an ordinary real number.

$$\lim_{x \rightarrow a} f(x) = \infty$$

- A vertical line $x = a$ is a **vertical asymptote** when the function becomes unbounded as x approaches a from at least one appropriate direction.
- For the following function the vertical asymptote is $x = 3$.

$$f(x) = \frac{2x}{x - 3}$$

- For the tangent function, the vertical asymptotes are

$$x = \frac{\pi}{2} + n\pi, \quad n \in \mathbb{Z}$$

- For the natural logarithm, the vertical asymptote is

$$x = 0$$

- The central idea of this topic is therefore that **limits describe local behavior**: they tell us what a function approaches, how it approaches from each side, and whether its behavior remains finite, oscillates, or becomes unbounded.

Exercises

Exercises Set 1

Solve the odd number exercises from exercise 1 to 51.

Exercise Set 2 (Reading Material)

Self study objective 4 (Infinite Limits; Vertical Asymptotes)

- Exercises help transform theoretical concepts into practical understanding.
- Mathematics is learned by doing and solving exercises will train you to analyze problems, select appropriate methods, and construct logical solutions.
- Attempting problems sometimes leads to mistakes which provide opportunities for learning and improvement.
- Regular practice increases speed, accuracy, and confidence.
- Exercises are given in the Exercises file and you are expected to solve them on your own.